

EXP8

April 10, 2026

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[2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split, StratifiedKFold,
    cross_val_score
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline

from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, roc_auc_score, roc_curve
)
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[22]: df = pd.read_csv("heart_data.csv")

X = df.drop("cholesterol", axis=1)
y = df["diagnosis"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42
)
```

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[24]: models = {
    "Logistic Regression": Pipeline([
        ("scaler", StandardScaler()),
        ("model", LogisticRegression())
    ]),

    "SVM (RBF)": Pipeline([
        ("scaler", StandardScaler()),
        ("model", SVC(kernel='rbf', probability=True))
    ]),
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    "Random Forest": RandomForestClassifier(
        n_estimators=100,
        random_state=42
    )
}

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[26]: results = []

for name, model in models.items():
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred)
    rec = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)
    auc = roc_auc_score(y_test, y_prob)

    results.append([name, acc, prec, rec, f1, auc])

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[28]: results_df = pd.DataFrame(results, columns=[
    "Model", "Accuracy", "Precision", "Recall", "F1 Score", "AUC-ROC"
])

print(results_df)

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	Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
0	Logistic Regression	1.0	1.0	1.0	1.0	1.0
1	SVM (RBF)	1.0	1.0	1.0	1.0	1.0
2	Random Forest	1.0	1.0	1.0	1.0	1.0

```

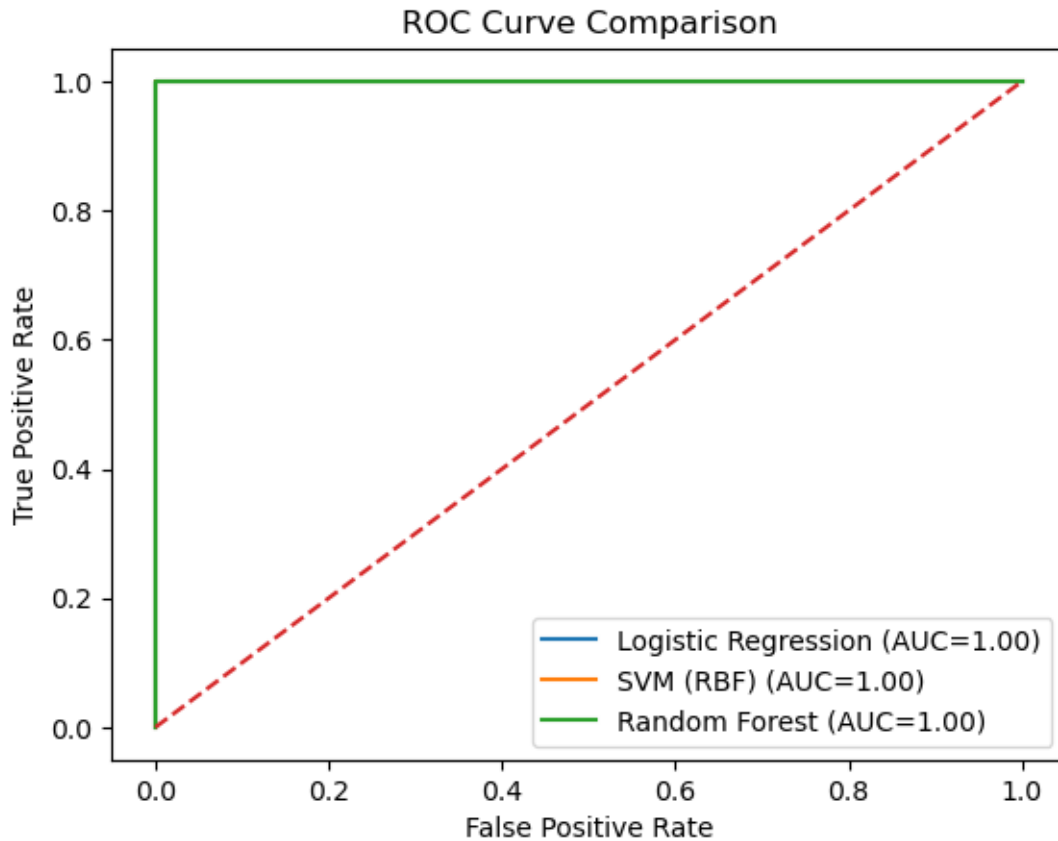
[30]: plt.figure()

for name, model in models.items():
    y_prob = model.predict_proba(X_test)[:, 1]
    fpr, tpr, _ = roc_curve(y_test, y_prob)
    auc = roc_auc_score(y_test, y_prob)

    plt.plot(fpr, tpr, label=f"{name} (AUC={auc:.2f})")

plt.plot([0,1], [0,1], linestyle='--') # random line
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve Comparison")
plt.legend()
plt.show()

```



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[32]: cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

cv_results = {}

for name, model in models.items():
    scores = cross_val_score(model, X, y, cv=cv, scoring='accuracy')
    cv_results[name] = scores.mean()

print("Cross-validation Accuracy:")
for k, v in cv_results.items():
    print(f"{k}: {v:.4f}")
```

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Cross-validation Accuracy:
Logistic Regression: 1.0000
SVM (RBF): 1.0000
Random Forest: 1.0000
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